



DEPARTMENT OF MATHEMATICS
NATIONAL INSTITUTE OF TECHNOLOGY KURUKSHETRA

B.Tech. Second Semester

END SEMESTER EXAMINATIONS

Date: May 09, 2025

Max. Marks: 50

Time: 02:00 pm – 05:00 pm

Paper: Integral Calculus and Difference Equations (MAIC 102)

Instructions:

- Answer all the questions.
- The calculator is not allowed into the examination hall.
- Start writing your answers from the first page. Do not leave it blank. No extra answer sheets will be provided.

1. Verify Stokes' theorem for $A = y^2\hat{i} + xy\hat{j} - xz\hat{k}$ where S is the hemisphere $x^2 + y^2 + z^2 = a^2, z \geq 0$. [7M]
2. Find the Fourier series of $f(x) = \begin{cases} 0 & : -\pi \leq x \leq 0 \\ x^2 & : 0 \leq x \leq \pi \end{cases}$, which is assumed to be periodic with period 2π . [7M]
3. Find the general solution of $(E^2 + 4)y_n = \cos(\alpha n)$. [7M]
4. Using Z-transform, solve the difference equation $u_{n+2} - 4u_{n+1} + 3u_n = 5^n$. [7M]
5. Find the Frobenius series solution about $x = 0$, of the equation $(1 - x^2)y'' - 2xy' + 6y = 0$. [7M]
6. Express $\int_0^1 x^m(1 - x^p)^n dx$ in terms of Beta function and hence evaluate the integral $\int_0^1 x^{3/2}(1 - \sqrt{x})^{1/2} dx$. [7M]
7. A solid fills the region between two concentric spheres of radii a and b , $0 < a < b$. The density at each point is inversely proportional to its square of distance from the origin. Find the total mass. [4M]
8. Find the directional derivative of the function $f = x^2 - y^2 + 2z^2$ at the point $P(1, 2, 3)$ in the direction of the line PQ where Q is the point $(5, 0, 4)$. [4M]